

Validation of the refinement

**Information from the Ist file
and other refinement details**

Start of the refinement

Reading the input file and establishing connectivity

Covalent radii and connectivity table for bfs79 in P-1

C 0.770
H 0.320
B 0.820
F 0.640
S 1.030

S1 - C19 C22 B1
B1 - C13 C1 C7 S1
C1 - C2 C6 B1
C2 - F1 C3 C1
C3 - F2 C4 C2
C4 - F3 C3 C5
C5 - F4 C4 C6
C6 - F5 C5 C1
C7 - C12 C8 B1
[...]
F1 - C2
F2 - C3
F3 - C4
[...]

We are refining atoms on specified position with spherical electron distribution. The connectivity information thus has no influence on the refinement. We might want to change it :

- Problems with AFIX, e. g. organometallic complexes
- Too many or missing bonds (coordination compounds)

CONN xx xxx xxx governs the connectivity list (see manual)

BIND atom1 atom2 adds a bond

FREE atom1 atom2 deletes a bond

The refinement – R_{sigma}

Reading the *.hkl file with the reflection intensities

example.lst:

20975 Reflections read, of which 685 rejected

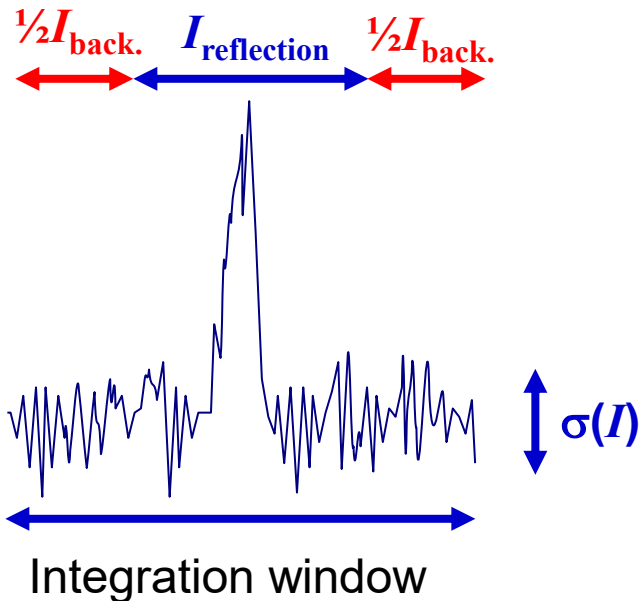
...

2847 Unique reflections, of which 0 suppressed

R(int) = 0.0275

R(sigma) = 0.0111

Friedel opposites merged



$$R_{\text{sigma}} = \frac{\sum \sigma(F_o^2)}{\sum F_o^2}$$

- Measure of the signal/noise ratio
- In first approximation, R1 (error of our model, see later) cannot be much better than R_{sigma}

The refinement - step by step

Reading the *.hkl file with the reflection intensities

example.lst:

h	k	l	Fo ²	Sigma	Why rejected
0	0	1	104.55	5.29	observed but should be systematically absent
2	0	1	74.15	6.04	observed but should be systematically absent

etc.

20975 Reflections read, of which 685 rejected

-10 ≤ h ≤ 10, -13 ≤ k ≤ 13, -18 ≤ l ≤ 18, Max. 2-theta = 137.96

23 Systematic absence violations

Inconsistent equivalents etc.

h	k	l	Fo ²	Sigma(Fo ²)	N	Esd of mean(Fo ²)
2	0	0	127128.20	518.46	4	3194.33
0	2	0	65677.31	243.31	6	1223.07

etc.

14 Inconsistent equivalents

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Rejected reflections:

- Systematic absences violations
- Inconsistent equivalents
- Reflections with $I < -3 \cdot \sigma(I)$

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The refinement – Merging of reflections

Reading the *.hkl file with the reflection intensities

Number of measured reflections

example.lst:

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-10 =< h =< 10, -13 =< k =< 13, -18 =< l =< 18, Max. 2-theta = 137.96
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- Number of **independent** measured reflections (neither repeated measurements, nor related by symmetry)
- Merging is ruled by the MERG command (see SHELXL manual). We need to change this only in special cases (twinning or merging of (meaningless) Friedel pairs)

The refinement - step by step

1. Reading of the *.hkl file

example.lst:

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0	0	1	104.55	5.2
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etc.

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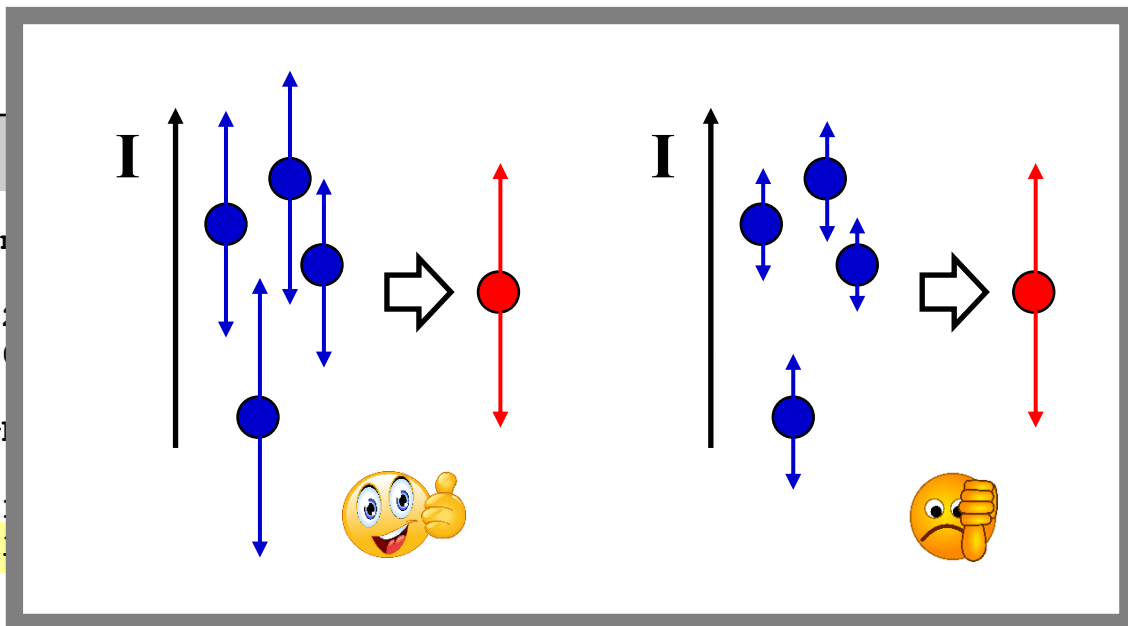
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Inconsistent equivalents:
 σ of averaging $\gg$ σ of measure

Rejected reflections:

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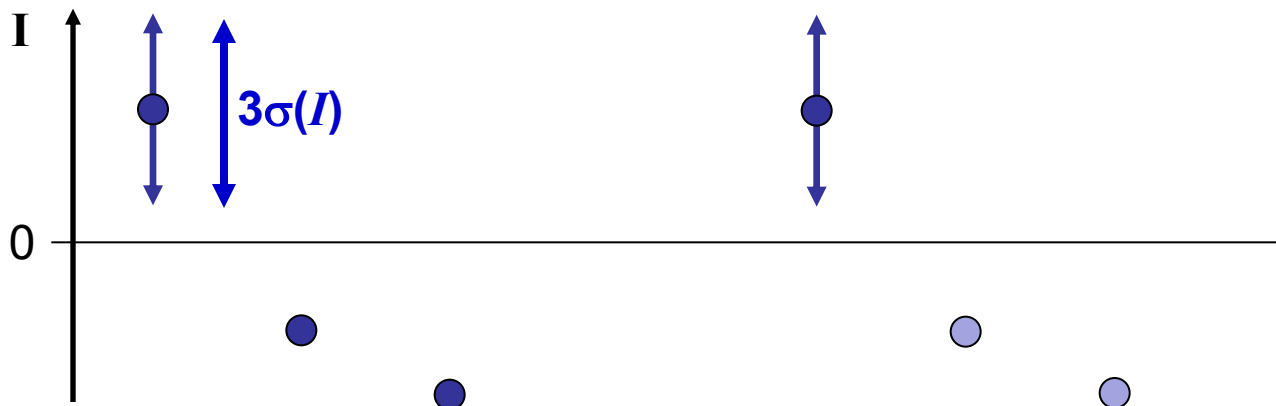
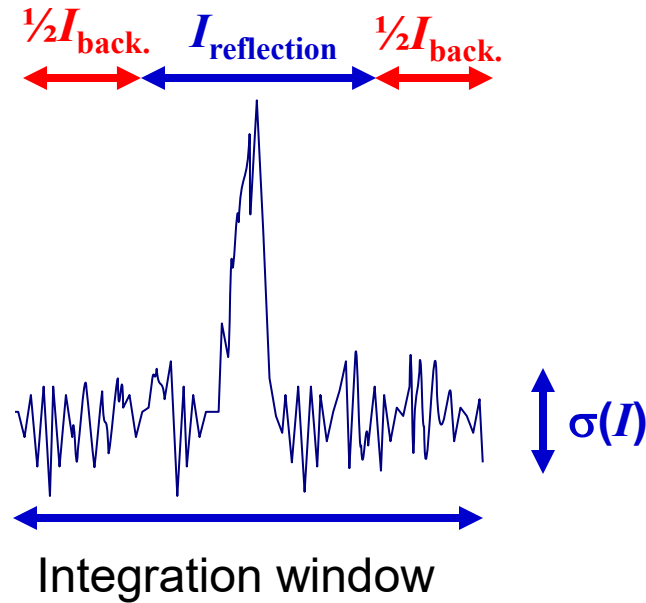
Rejected reflections:

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- Inconsistent equivalents
- Reflections with $I < -3 \cdot \sigma(I)$

Why use reflections with negative intensity?

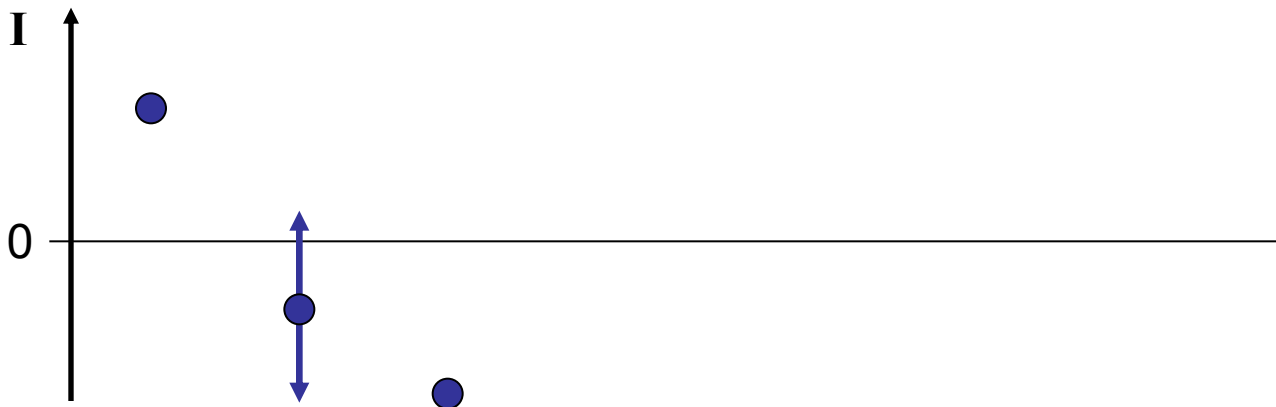
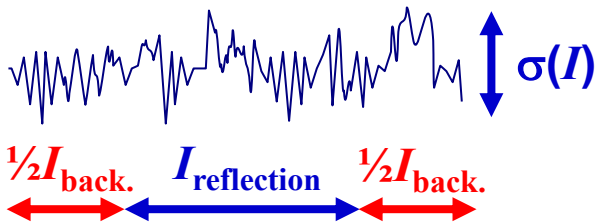
- During the integration of the reflections, their intensity is calculated by

$$I = I_{\text{reflection}} - I_{\text{background}}$$



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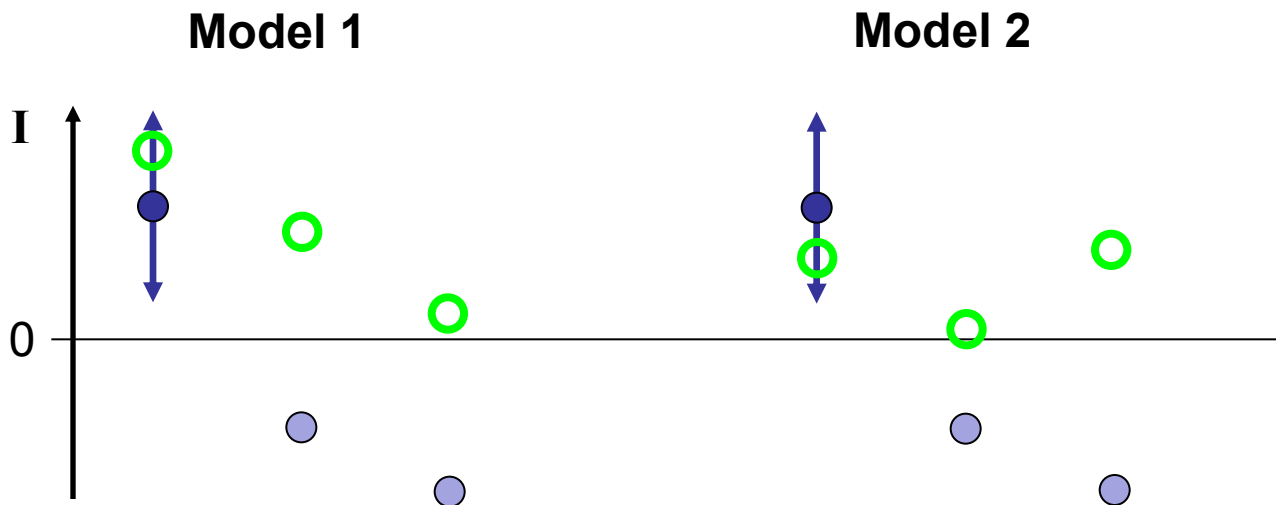
- During the integration of the reflections, their intensity is calculated by $I = I_{\text{reflection}} - I_{\text{background}}$
- For weak reflections, it can happen that the result is negative, due to a large peak in the noise of the background.



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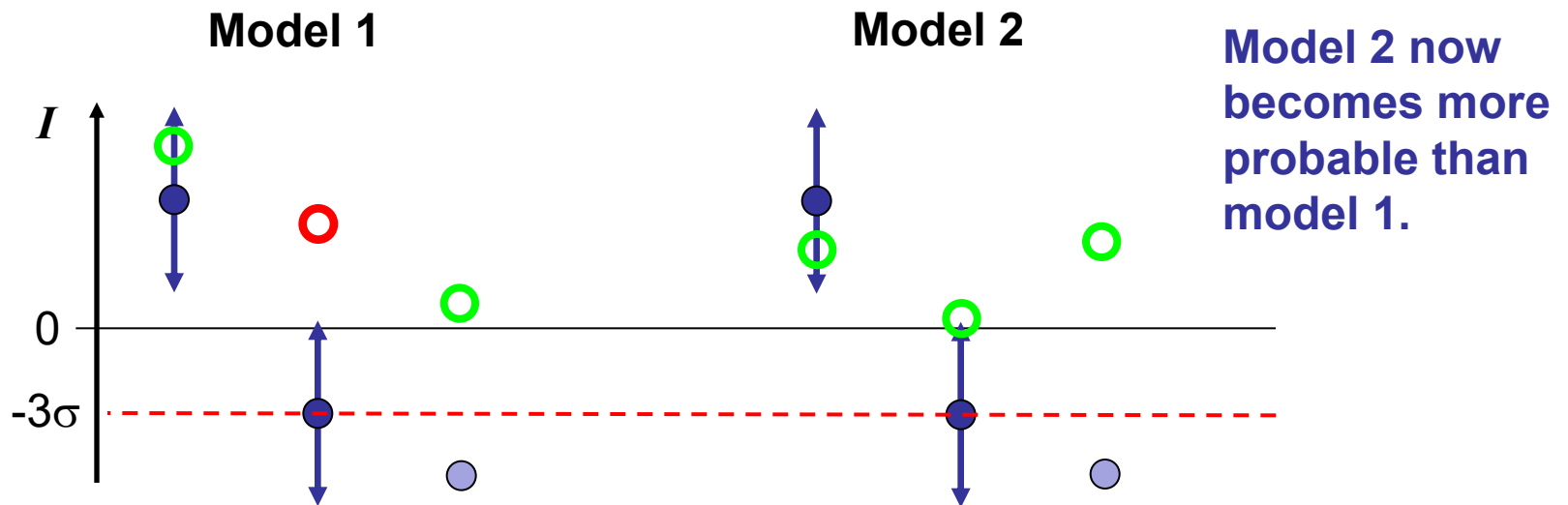
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- For weak reflections, it can happen that the result is negative, due to a large peak in the noise of the background.

If we consider only reflections with $I > 0$, the two models 1 and 2 have the same probability / the same agreement with the experimental data.



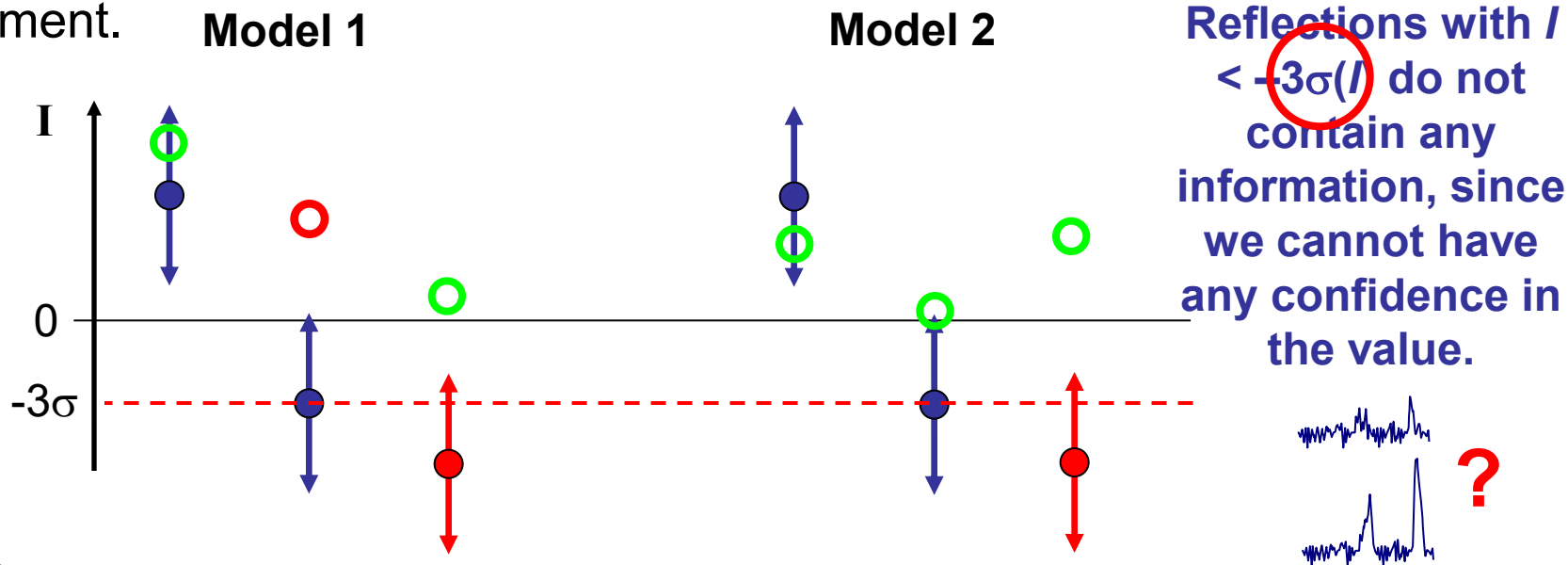
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- For weak reflections, it can happen that the result is negative, due to a large peak in the noise of the background.
- Reflections with $0 < I < -3\sigma(I)$ contain important information, since $I = 0$ is in the range of the experimental error.



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- During the integration of the reflections, their intensity is calculated by $I = I_{\text{reflection}} - I_{\text{background}}$
- For weak reflections, it can happen that the result is negative, due to a large peak in the noise of the background.
- Reflections with $0 < I < -3\sigma(I)$ contain important information, since $I = 0$ is in the range of the experimental error.
- Reflections with $I < -3\sigma(I)$ are impossible, since all possible values in the error range are negative. Thus, there was a problem with the determination of this reflection, and **it needs** to be excluded from the refinement.

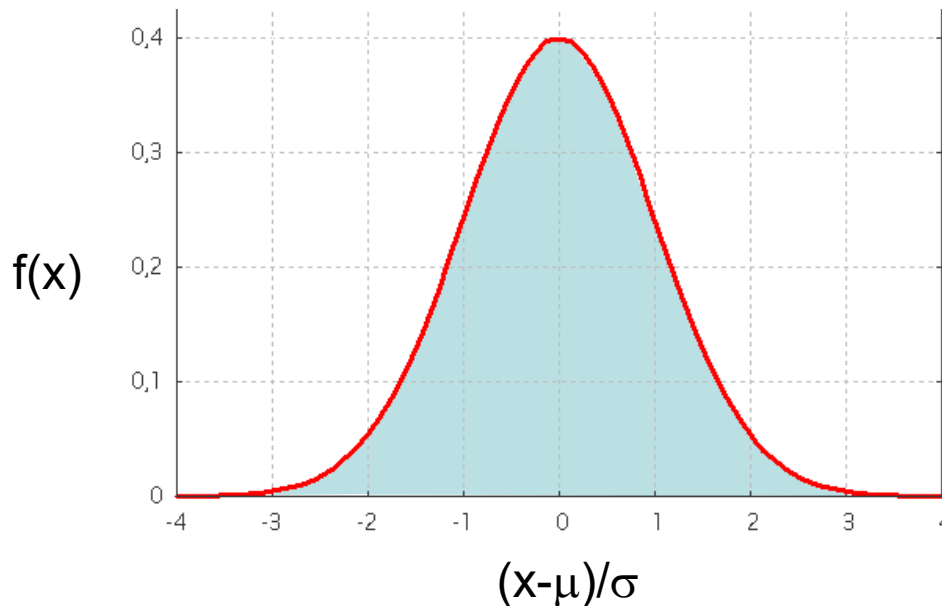


3σ : Why always $\pm 3\sigma$?

An arbitrary distribution of errors has a Gaussian distribution.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{(x-\mu)}{\sigma}\right)^2}$$

μ : correct value
 σ : standard deviation



$$\int_{\mu-\sigma}^{\mu+\sigma} f(x) = 0.68 = 68\%$$

$$\int_{\mu-2\sigma}^{\mu+2\sigma} f(x) = 0.96 = 96\%$$

$$\int_{\mu-3\sigma}^{\mu+3\sigma} f(x) = 0.99 = 99\%$$

99.7% probability that x is found in the interval $\mu \pm 3\sigma$

$$a = 1.234(5): a=1.234; \sigma=0.005 \rightarrow a = 1.234 \pm 0.015$$

The refinement – «Merging» – R_{int}

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R(sigma) = 0.0111

Friedel opposites merged

$$R_{\text{int}} = \frac{\sum |F_o^2 - F_o^2(\text{mean})|}{\sum F_o^2}$$

R_{int} : Error of merging (measure of precision / reproducibility of the measurement)

Possible reasons for high R_{int} (R_{int} is high if $R_{\text{int}} > 2 \cdot 3 \cdot R_{\text{sigma}}$):

- Incorrect Laue group (crystal system)
- Missing or wrong absorption correction
- Crystal decomposition
- Twinning
- Goniometer problems (hidden reflections, bad alignment, etc.)

Correlation matrix elements

test.lst:

Largest correlation matrix elements

0.853 U11 Fe1 / OSF

0.728 U11 S2 / U11 Fe1

0.524 U11 S1 / U11 Fe1

0.771 U11 S2 / OSF

0.588 U11 S1 / OSF

0.543 U12 S1 / U23 S1

The **correlation matrix**, X , describes the correlation (interdependence) between the variables in a refinement. The correlation of a variable with itself, X_{ii} , is always 1. The correlation between two variables, X_{ij} , is 1 (or -1) if they are linearly dependant and zero if they are independent.

The **covariance matrix** is essentially the same, but the correlation matrix is unit-free and scaled so that $X_{ij} = -1 \dots 1$.

There is always some interdependency between variables. **Values in the correlation matrix > 0.5 , however, indicate significant correlation between variables and need to be inspected carefully.** They might indicate missed symmetry (or other problems due to overparametrization).

Correlation matrix elements

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0.543 U12 S1 / U23 S1

Values in the correlation matrix > 0.5 , however, indicate significant correlation between variables and need to be inspected carefully.

Some correlations are, however, anticipated and is no reason to worry:

.ins/.res:

FVAR 0.81445

- **Correlations between the ADPs of heavy atoms and the OSF or between the ADPs of heavy atoms** : A large quantity (if not the majority) of the electrons is concentrated in the heavy atoms. A diminution of the ADPs (=smaller integration area) can thus be compensated by an increase of the overall scale factor (OSF).
- **Correlation between ADPs of the same atom** : A rotation of 90° of the ADP ellipsoid around axis 1, transforms U_{22} into U_{33} and vice versa. A certain correlation between ADPs is thus normal, in particular if the ADP are relatively isotropic.

Correlation matrix elements

test.lst:

Largest correlation matrix elements

0.661 z C11 / z C61	0.650 x C7 / x C57	0.638 x C20 / x C70
0.660 y C12 / y C62	0.648 x C2 / x C52	0.636 z C3 / z C53
0.659 x C14 / x C64	0.647 x C6 / x C56	0.633 x C8 / x C58
0.657 x C11 / x C61	0.647 z C6 / z C56	0.633 y C11 / y C61
0.655 x C15 / x C65	0.647 x C16 / x C66	0.632 x C12 / x C62
0.654 x C17 / x C67	0.647 z C10 / z C60	0.631 x F11 / x F61
0.653 z C15 / z C65	-0.640 U11 C16 / U11 C66	0.631 z C9 / z C59
0.652 x C10 / x C60	0.639 z C12 / z C62	0.630 x C18 / x C68

Correlations between the atomic position of independent atoms or between ADPs of light atoms typically indicate a problem. In most cases, a missed symmetry (= wrong space group).

- Verify manually which atoms are correlated. Is there a symmetry operation which could possibly correlate all these atoms?
- Check for other possible space groups?
- Use PLATON/ADDSYM
- Was another, more symmetric unit cell, offered during integration?

Confidence factors

Final Structure Factor Calculation for vivan6 in P2(1)/c

Total number of l.s. parameters = 227 Maximum vector length = 623 Memory required = 49 / 2946 / 29281

wR2 = 0.0800 before cycle 6 for 2847 data and 0 / 227 parameters

GooF = S = 1.041; Restrained GooF = 1.041 for 0 restraints

Weight = $1 / [\sigma^2(F_o^2) + (0.0457 * P)^2 + 0.57 * P]$ where $P = (\text{Max}(F_o^2, 0) + 2 * F_c^2) / 3$

wR2 = 0.0800, GooF = S = 1.041, Restrained GooF = 1.041 **for all data**
R1 = 0.0307 for 2752 $F_o > 4\sigma(F_o)$ and 0.0312 for all 2847 data

SHELXL minimises M, the difference of observed and calculated intensity:

$$M = \sum w (F_o^2 - F_c^2)^2$$

(Thus the smaller is M, the better is the agreement between our model and the experimental data.)

Confidence factors

SHELXL minimises:

$$M = \sum w(F_o^2 - F_c^2)^2$$

M increases with the number of reflections and their intensity. M is thus useful to optimise the structure, but not to compare the quality of the structure for two different experiment or to judge the quality of the model. We require thus a value independent from the number and intensity of the reflections.

Confidence factor, residual, R factor:

R2 →

$$wR_2 = R_w(F^2) = \sqrt{\frac{\sum w(F_o^2 - F_c^2)^2}{\sum w(F_o^2)^2}}$$

For purely statistical reasons, refinement against F^2 results in R factors twice as high than refinement against F . To facilitate comparisons (and to increase acceptance of the new refinement against F^2), SHELXL also calculates a factor R (or R1) based on F .

$$R_1 = \frac{\sum ||F_o| - |F_c||}{\sum |F_o|} \quad \leftarrow \textbf{R1}$$

R1 cannot be calculated with the weighing scheme optimized for F^2 .

*Traditionally, it should also be calculated **only for reflections with $I > 2s(I)$** .*

Goodness of Fit

The GoF or GooF is another value describing the quality of our model:

$$GooF = S = \sqrt{\frac{\sum w(F_o^2 - F_c^2)^2}{N_{Ref.} - N_{Par.}}}$$

$N_{Ref.}$: number of independent reflections, $N_{Par.}$: number of parameters

Contrary to the R factor, which also depends on the noise, S is relatively independent of the noise. S, despite its way of calculation above, is the ratio of expected vs. observed deviations (errors).

S should have a value of appr. 1.

$S > 1$: errors larger than expected. Bad model or bad data/parameter ratio

$S < 1$: errors are smaller than expected. Typically indicates incorrect $\sigma(I)$: problems with the absorption correction or the space group.

Criteria for a good structure

SHELXL calculates 4 confidence factors :

- **wR2 (all data)**
- **wR2 (observed data, $I > 2\sigma(I)$)**
- **R1 (all data)**
- **R1 (observed data, $I > 2\sigma(I)$)**

Refinement against F^2 requires an optimized weighing scheme, but this scheme cannot be used for calculation of R1.

The important factors are **wR2 (all data)** (since all reflections are used in refinement) and **R1 (observed data)**, for comparison purposes, since typically only reflections with $F < 3\sigma(F)$, above the observation threshold, were used in refinements against F . ($F < 3\sigma(F) \approx I < 2\sigma(I)$).

	Good	Acceptable	Problematic	Really problematic
R1	< 5%	< 7%	>10%	>15%
wR2	< 12%	< 18%	>25% (or » 2*R1)	>35%
S	0.9-1.1	0.8-1.5	<0.8 or >2	<0.6 or >4

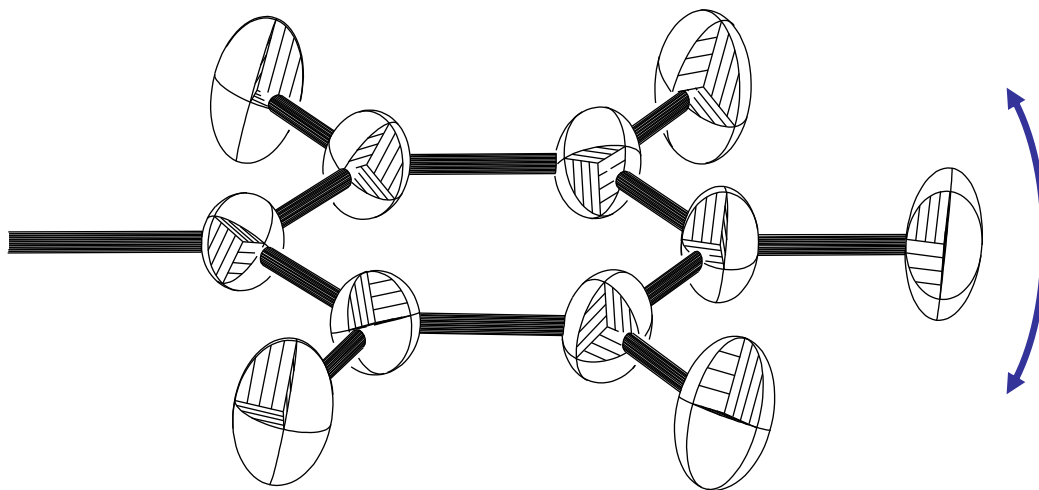
These ranges are just providing a measuring stick. There are completely fine, publishable structures with R1=17% and unpublishable ... stuff with R=4%. It depends on the *reason* for the bad values.

Thermal parameters

With the exception of wrong space group assignment, nearly all other problems of a structure refinement affect the thermal ellipsoids stronger than atom positions. For weak/bad structures, the thermal parameters thus become the garbage bin of the refinement. (Which is why crystallographers want to show ADPs with 50% probability while chemists prefer ball-and-stick pictures or smaller ellipsoids.)

In general :

- ADPs (= thermal movement) should be similar for similar atoms.
- The direction and size of the thermal ellipsoid should be in agreement with the lowest-energy vibration expected.



Manual inspection of thermal parameters

*.lst:

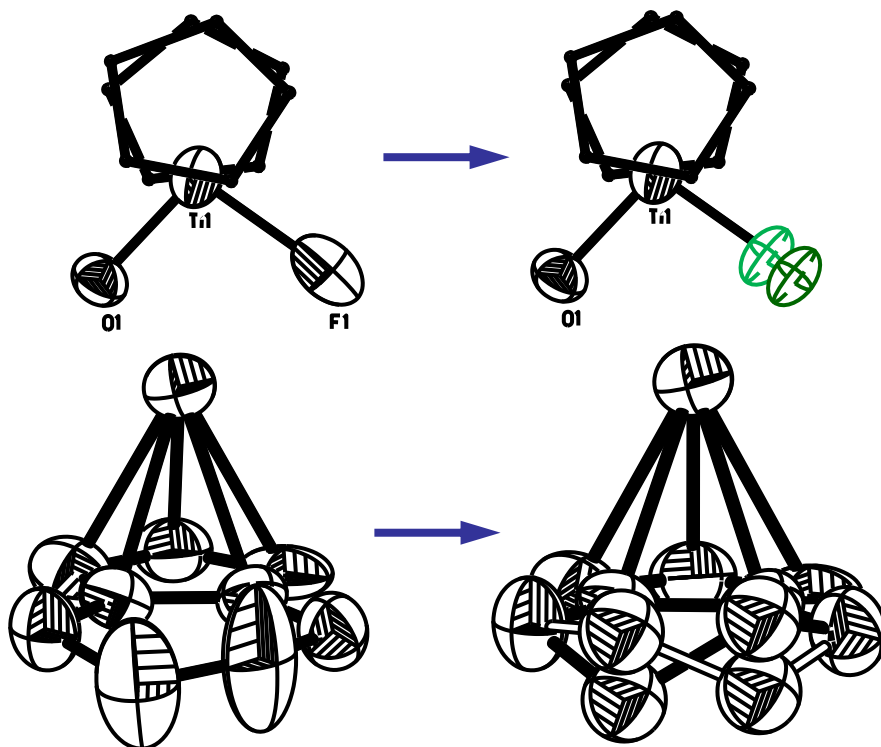
Principal mean square atomic displacements U

[...]

0.3098 0.0893 0.0464 C4 **may be split** into 0.6218 0.2673 0.2408 and 0.6118 0.2471 0.2666

0.3100 0.0924 0.0392 C5 **may be split** into 0.5976 0.3191 0.3424 and 0.5834 0.3017 0.3597

Elongated ADPs often suggest unrefined disorder



Of course this has to be confirmed by manual inspection

Most Disagreeable Reflections

Most Disagreeable Reflections (* if suppressed or used for R_{free}).

Error/esd is calculated as $\sqrt{wD^2/\langle wD^2 \rangle}$ where w is given by the weight formula, $D = F_o^2 - F_c^2$ and $\langle \rangle$ refers to the average over all reflections.

h	k	l	F _o ²	F _c ²	Error/esd	F _c /F _c (max)	Resolution(Å)
-1	0	4	34915.33	49953.90	6.93	1.000	3.80
0	5	4	44.72	18.59	6.14	0.019	1.96
-1	5	5	12.66	36.45	5.55	0.027	1.84
3	5	1	89.93	52.46	5.42	0.032	1.75
0	6	1	104.57	68.97	4.51	0.037	1.90
2	3	3	7.90	0.95	4.49	0.004	2.34
[...]							

Regardless how good your dataset, one reflection has to show the biggest deviation between calculated and observed intensity. **Most** disagreeable does not have to mean **very** disagreeable.

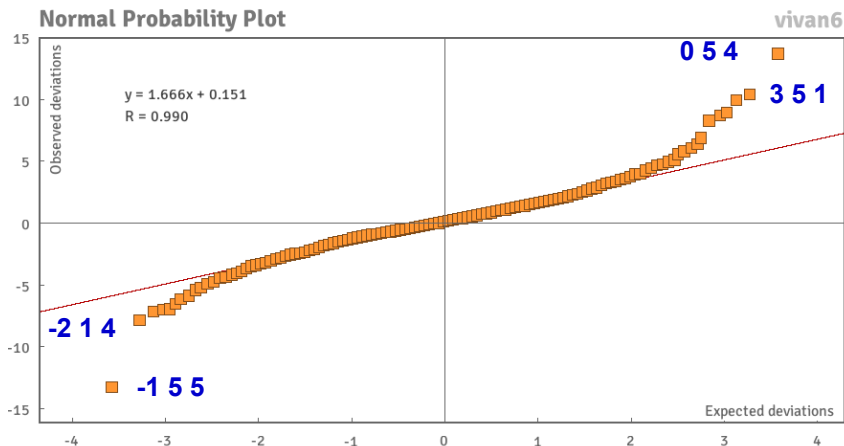
Look out for :

- A small number of reflections with much higher (at least 2x) the error of the others.
- Systematic tendencies (h = 3n, all high resolution, all low resolution...)

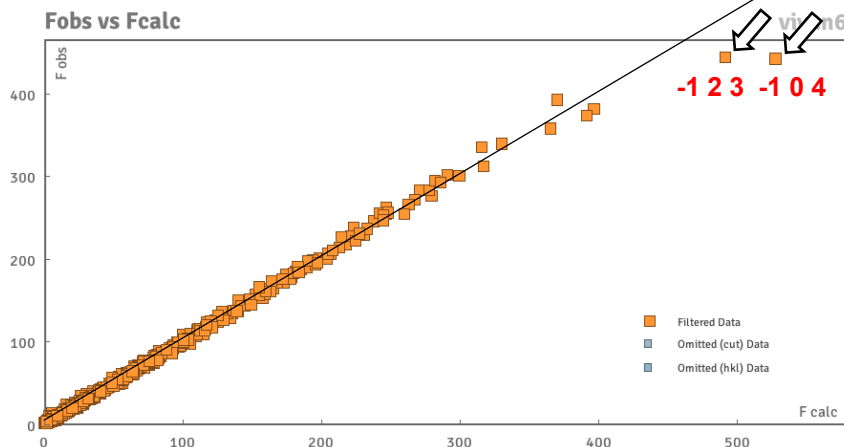
Most Disagreeable Reflections – OLEX2

The same list is available in OLEX2 under INFO-Bad reflections. This list is good to spot systematic tendencies (h=3n). Other systematic errors are better spotted with the statistics plots in OLEX2.

Normal Probability Plot



Fobs vs Fcalc



Home Work View Tools Info

Recent Files

Electron Density Peaks

Refinement Indicators

Bad Reflections

OMIT all reflections where |Error/esd| > than 10 ClearOMITs

Unsigned sorting by Error/esd (refine to update) ☒

	H	K	L	Error/esd	d/A	
→	-1	0	4	6.93	3.80	omit Edit...
→	0	5	4	-6.14	1.96	omit Edit...
→	-1	5	5	5.55	1.84	omit Edit...
→	3	5	1	-5.42	1.75	omit Edit...
	0	6	1	-4.51	1.90	omit Edit...
	2	3	3	-4.49	2.34	omit Edit...
	1	6	1	-4.35	1.84	omit Edit...
	-3	2	3	4.30	2.52	omit Edit...
→	-1	2	3	4.20	3.72	omit Edit...

Click to omit reflections from refinement

*.ins

OMIT -1 2 3

OMIT -1 0 4



R1 :

3.07% → 2.95%

However, always check if you can find the **reason** for this deviation (extinction, absorption correction, etc.) and correct it!

Residual electron density

*.lst

Electron density synthesis with coefficients Fo-Fc

Highest peak **0.20** at 0.4190 0.6862 0.1604 [0.69 Å from N3]

Deepest hole **-0.33** at 0.7000 0.3938 0.1491 [0.38 Å from P1]

If our model is correct, we should have correctly described all electron density. Thus the remaining electron density ($\rho - \rho_{\text{modèle}}$) should be zero. However, we still get notable maxima due to remaining noise, electron density in bonds, Fourier truncation errors, insufficient description of disorder, etc.

Acceptable values for remaining electron density :

- Do look also at the biggest hole! If electron density is simply due to remaining noise, the whole should be as big as the maximum.
- For light-atom structures (H – F) : **< 0.5 e⁻/Å³**
- For structures with heavy atoms : up to 10% of the electron density of the heavy atom (although rarely that high) can be found in a distance of around 1 Å of the heavy atom (Fourier truncation errors).
- If nothing can be done and there is no “good reason”, it has to be explained (comment in CIF). ***This lowers the reliability of your data!***

Fourier truncation errors

Fourier truncation errors appear if the number of terms of the infinite sum of the Fourier transformation is limited. They appear as “wiggles” in places where electron density changes drastically.

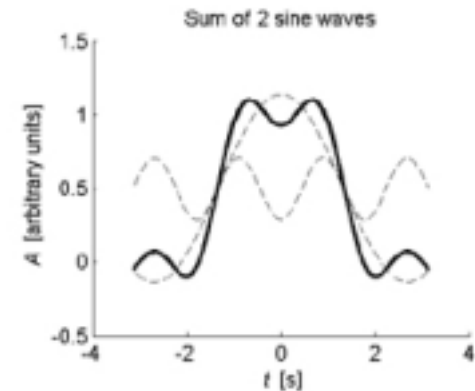
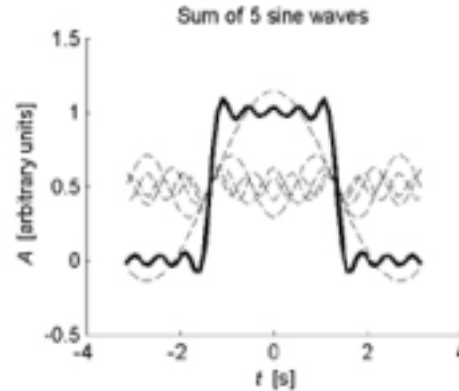
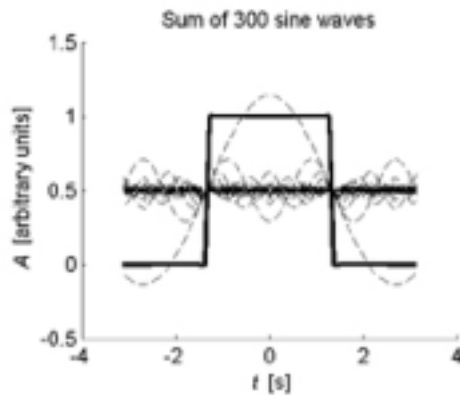
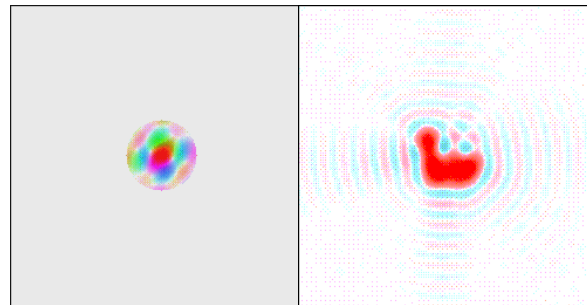
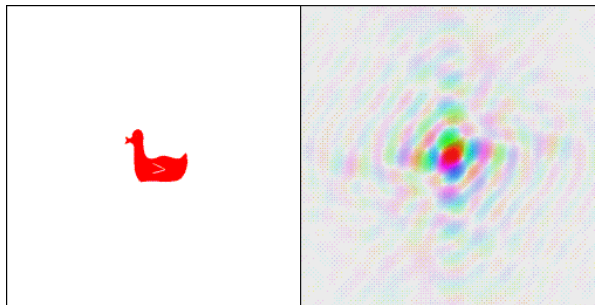
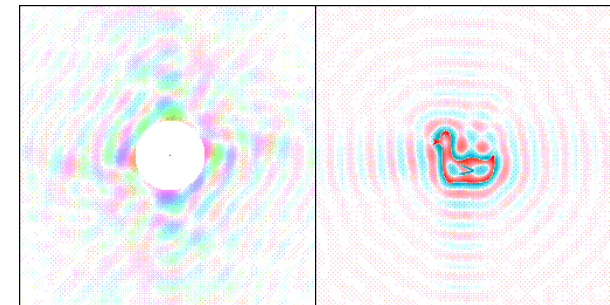


Image: <http://www.theses.ulaval.ca/2005/23016/apd.html>

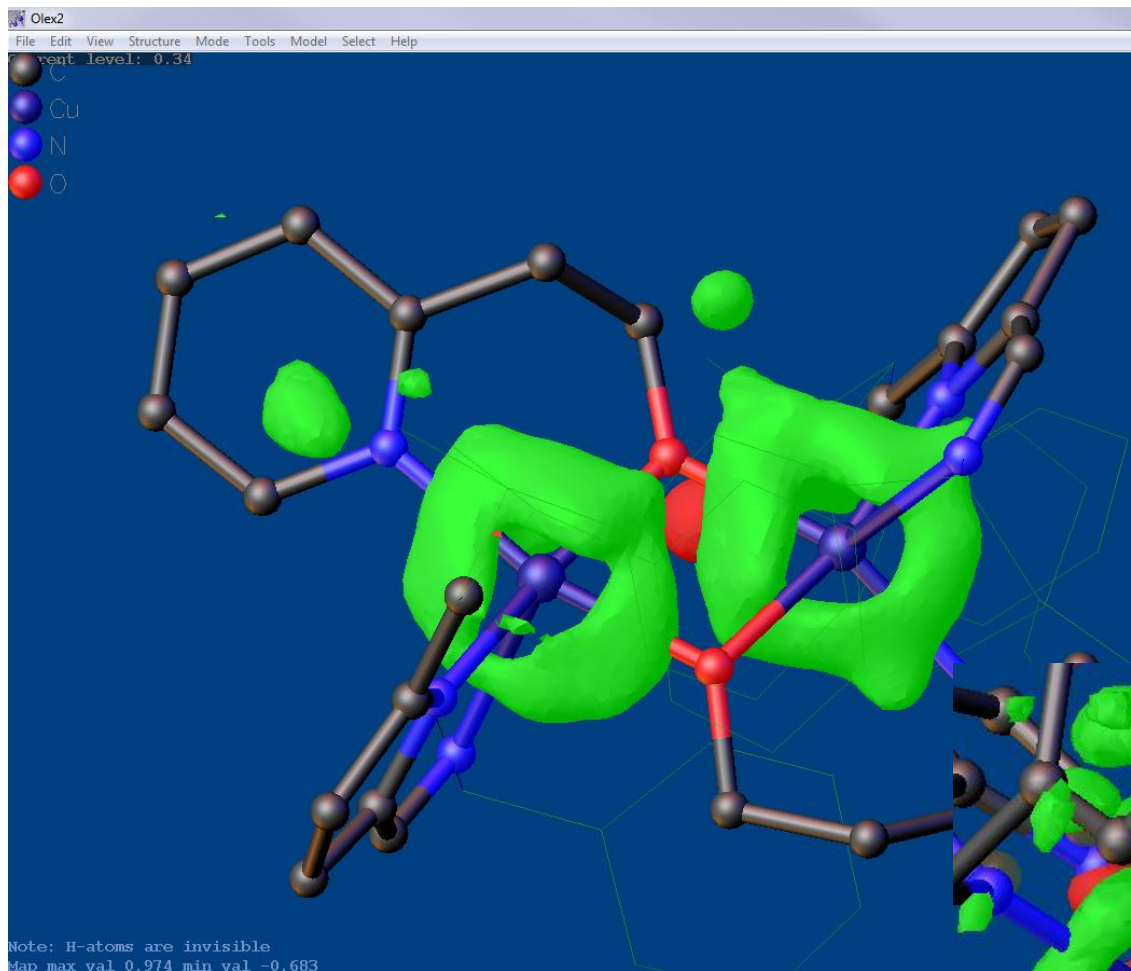


High θ suppressed



Low θ suppressed

Fourier truncation errors



Home Work View **Tools** Info

HART

AutoChem 3.1

Images

Maps

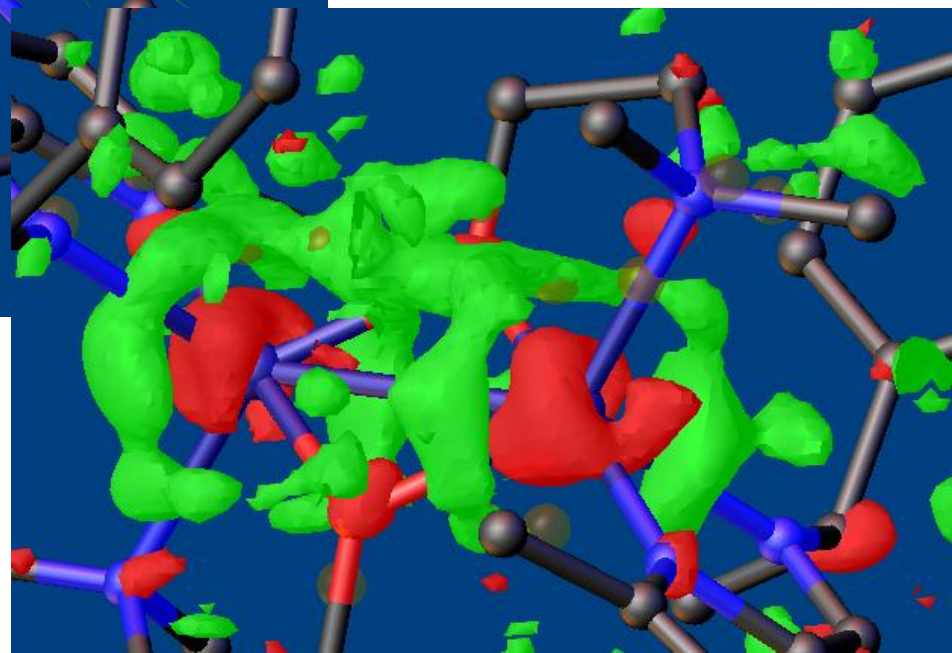
View surface min: -0.683 max: 0.974 Extended ☐ Edit

Level /eA⁻³ -0.34

Electron Density

Map diff Source olex Res/Å 0.1 ☒ Mask Hide Map

The image shows the Olex2 software interface, specifically the Tools menu. The Maps panel is open, showing the current map type (diff), source (olex), resolution (0.1 Å), and various options like Mask and Hide Map. The current level is set to -0.34 eA⁻³. Red arrows point to the Tools menu, the Level input field, and the Hide Map button.



How to find and what to do with residual electron density

OLEX2 : info Q1

test.cif :

_refine_diff_density_max 0.425
_refine_diff_density_min -0.560

test.lst :

Electron density synthesis with coefficients Fo-Fc
Highest peak 0.42 at 0.2140 0.0398 0.5135 [0.40 Å from C2]
Deepest hole -0.56 at 0.1424 0.4014 0.6865 [0.53 Å from F3]

Possible reasons and consequences for $\rho > 0.5 \text{ e}^-/\text{\AA}^3$

- Missing/misassigned hydrogen atoms → Correct the problem.
- Fourier truncation errors → If close (appr. 1 Å) to a heavy atom or on a special position: explain in CIF. If not, it's not a Fourier truncation error.
- Bad absorption correction → Correct the problem.
- Disorder → Correct the problem. If not possible, explain in CIF.
- Twinning → Correct the problem.
- Electron density in bonds → Should not be $> 0.5 \text{ e}^-/\text{\AA}^3$
- Unknown → Explain in CIF why you can exclude all of the above and admit not knowing the reason. This, of course, reduces the quality of the structure.

Be aware of OLEX2 warnings

BFS79 $P\bar{1}$
D:\presentations\cccw23\BFS79\nicely numbered\BFS79.res

C22H8BF15S

a = 9.5892(8)	$\alpha = 111.904(6)^\circ$	Z = 2		R_1	3.11 %		
b = 10.8264(10)	$\beta = 96.814(7)^\circ$	Z' = 1					
c = 11.3959(9)	$\gamma = 100.727(7)^\circ$	V = 1055.29(17)					
d min (Mo)	0.78	I/ σ (I)	27.0	Rint	3.42%	Full 50.5°	97.4
2 θ =54.0°				m=1.77		95% to 54.0°	
Shift	0.002	Max Peak	0.3	Min Peak	-0.2	GooF	1.020

C27H28CuN3O4S

a = 11.0340(3)	$\alpha = 90^\circ$	Z = 4		R_1	3.69 %		
b = 15.4879(5)	$\beta = 90^\circ$	Z' = 1					
c = 17.0096(5)	$\gamma = 90^\circ$	V = 2906.83(15)					
d min (Ga)	0.77	I/ σ (I)	41.7	Rint	5.81%	Full 107.2°	99.8
2 θ =121.5°				m=13.74			
Shift	0.195	Max Peak	0.7	Min Peak	-0.3	GooF	1.141
						Hooft	.4336(9)

TWIN LAW (-1.0, 0.0, 0.0, 0.0, -1.0, 0.0, 0.0, 0.0, -1.0), BASF [0.433(4)]

C68H120N8

a = 13.108(6)	$\alpha = 90^\circ$	Z = 1		R_1	5.50 %		
b = 13.108(6)	$\beta = 90^\circ$	Z' = 0.125					
c = 9.526(4)	$\gamma = 90^\circ$	V = 1636.8(13)					
d min (Cu/a)	0.83	I/ σ (I)	10.7	Rint	14.44%	Full 135.3°	99.6
2 θ =135.3°				m=10.01			
Shift	0.000	Max Peak	0.1	Min Peak	-0.2	GooF	0.884

TWIN LAW (-1.0, 0.0, 0.0, 0.0, -1.0, 0.0, 0.0, 0.0, -1.0), BASF [0.5(19)]

Structure said19 loaded

fate032 $P2_1/c$
D:\X-ray\structures\fate032\hein\fate032.res

C55H60N4O4Zr

a = 13.8575(3)	$\alpha = 90^\circ$	Z = 8		R_1	15.48 %		
b = 47.9452(16)	$\beta = 103.216(1)^\circ$	Z' = 2					
c = 14.6072(4)	$\gamma = 90^\circ$	V = 9448.0(5)					
d min (Cu/a)	0.82	I/ σ (I)	16.7	Rint	8.86%	Full 135.4°	99.9
2 θ =140.2°				m=5.68			
Shift	0.001	Max Peak	4.5	Min Peak	-2.0	GooF	1.180

Structure fate032 loaded

Home Work View Tools Info

Solve Refine Draw Report

Program SheXL L.S. Cycles 20 Peaks 13

hkl file fate032.hkl Mon Aug 23 21:35:35 2021

Weight ☒ .042 | .041 255 | 257 EXTI ☐ SWAT ☐ ACTA

a = 10.2556(10)	$\alpha = 74.418(4)^\circ$	Z = 4		R_1	5.41 %		
b = 18.7318(18)	$\beta = 88.915(4)^\circ$	Z' = 2					
c = 20.538(2)	$\gamma = 82.638(4)^\circ$	V = 3768.6(7)					
d min (Cu/a)	0.91	I/ σ (I)	7.0	Rint	11.20%	Full 114.9°	97.4
2 θ =114.9°				m=6.11			
Shift	0.001	Max Peak	0.3	Min Peak	-0.3	GooF	0.856

And have fun !

Reflection Statistics

hkl File matt901_0m_4.hkl Write graphics file

Normal Probability

-- Please Select --

- Wilson Plot
- Cumulative Intensity
- Systematic Absences
- Fobs-Fcalc
- lobs-lcalc
- I/sigma vs resolution
- cc_half_vs_resolution
- Rmerge vs resolution
- Fobs over Fcalc
- Completeness
- Normal Probability